BENCHMARK OF BEAM DYNAMICS CODE DYNAC USING THE ESS PROTON LINAC

EUROPEAN SPALLATION SOURCE

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OVERVIEW

This paper reports on:

- the SUCCESSFUL BENCHMARK OF DYNAC against TraceWin using the European Spallation Source (ESS) Proton Linac

- the ACCURACY AND EFFICIENCY of DYNAC in view of online modeling

DYNAC CODE IS MATURE

Originally authored by LAPOSTOLLE, TANKE, VALERO in the late 1980s

Modeled CERN LINAC2 & LINAC3 and SNS LINAC (benchmark against PARMILA, LORAS); successfully used for commissioning of CERN LINAC3

Benchmarked against Track and Impact (THPB065@LINAC12) for multi-charge state FRIB linac

DYNAC is being used for modeling of Re-Accelerator at MSU

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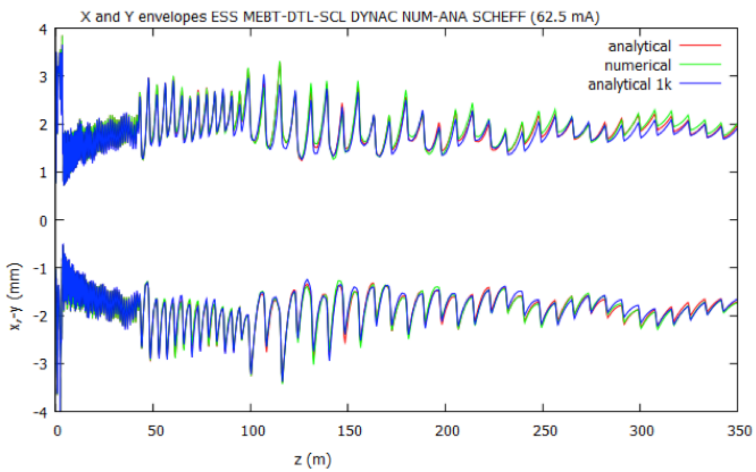
DYNAC IS ACCURATE

DYNAC contains:

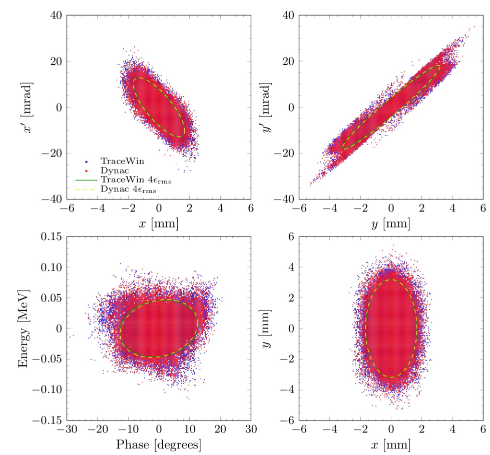
- **3 space charge routines** including a 2D (modified SCHEFF) and a 3D version (HERSC)
- **TWO COMPLETELY INDEPENDENT BEAM DYNAMICS METHODS FOR CAVITIES**:
Numerical (FAST) and Analytical (very FAST)

Below are examples of consistency within the code (left) and with TraceWin (right)

ANALYTICAL METHOD
(50k and 1k macro-particles)
&
NUMERICAL METHOD



DYNAC & TRACEWIN



DYNAC IS EFFICIENT

Computer: MAC, Intel Core i7,1600MHz DDR

ESS END2END SIMULATION (50k macro-particles)
in
350 sec with numerical method
54 sec with analytical method

DYNAC END2END SIMULATION in 2.5 s
with 1000 macro-particles

<u>TYPICAL USE IN COMMISSIONING IS SECTION BY SECTION (i.e. will be faster)</u>

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FINAL REMARKS

ACCURACY - the differences with other codes, when observed, are within acceptable limits and much lower than the expected measurement errors in commissioning

EFFICIENCY - the execution time is very well suited for online modelling

DYNAC IS OPEN SOURCE PROJECT

http://dynac.web.cern.ch/dynac/dynac.html

and same source can be used on Windows, MAC and Linux

(including graphics)

MORE DETAILS AT POSTER THPP043